Deep Learning-Based Seismic Damage Assessment of RC Frame Structures Using Acceleration Response Features

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Abstract

This study presents a machine learning framework for rapid seismic damage assessment of reinforced concrete (RC) frame structures using acceleration response features. Three classification models were compared: Random Forest (RF), Gradient Boosting (GBR), and Deep Neural Network (DNN). Twenty input features were extracted from structural response signals, including peak ground motion parameters, spectral accelerations, inter-story drift ratios, and structural characteristics. The models classify damage into four states: None, Slight, Moderate, and Severe. Using 5-fold stratified cross-validation on 2,000 synthetic response records, RF achieved the highest classification accuracy (98.2%) and weighted F1 score (0.9825). Feature importance analysis revealed that inter-story drift ratios and Arias intensity are the most discriminative features for damage classification. The proposed framework enables near-real-time damage assessment following earthquakes, supporting emergency response decision-making. The results demonstrate that ML-based approaches can provide accurate, consistent, and rapid structural damage evaluation, complementing traditional visual inspection methods and advancing automated post-earthquake safety assessment.

Keywords: seismic damage assessment; deep learning; RC frame; classification; inter-story drift; SHM; random forest

Highlights

• ML classifiers achieve over 97% accuracy for four-class seismic damage assessment

• Random Forest outperforms Gradient Boosting and DNN for tabular response features

• Inter-story drift ratios are the most discriminative features for damage classification

• Energy-based intensity measures outperform peak ground motion parameters

• Framework enables near-real-time post-earthquake building safety evaluation

INTRODUCTION

Rapid assessment of structural damage following earthquakes is critical for effective emergency response and resource allocation. Every year, seismic events cause significant casualties and economic losses worldwide, with reinforced concrete (RC) frame structures representing a large portion of the affected building stock in urban areas. In the immediate aftermath of an earthquake, emergency management authorities must quickly determine which buildings are safe to occupy, which require restricted access, and which pose an imminent collapse risk. This triage process directly impacts the safety of occupants and the efficiency of search-and-rescue operations.

Traditional post-earthquake damage assessment relies on visual inspection by trained engineers using standardized procedures such as ATC-20 in the United States or the European Macroseismic Scale (EMS-98). While these procedures provide a systematic framework, they are inherently time-consuming, subjective, and potentially dangerous in the immediate aftermath of a seismic event when aftershock hazard remains elevated. Field experience from major earthquakes, including the 2011 Christchurch earthquake and the 2023

Turkey-Syria earthquake sequence, has repeatedly demonstrated that the demand for rapid building assessment far exceeds the available inspection capacity, often requiring weeks or months to complete comprehensive assessments of affected areas.

The development of automated, data-driven damage classification systems has the potential to significantly accelerate the assessment process and improve the consistency of damage evaluations. The seismic performance assessment of RC frame structures has been extensively studied within the framework of performance-based earthquake engineering (PBEE), as described by Moehle (2015). The PBEE framework establishes formal relationships between ground motion intensity measures, engineering demand parameters (such as inter-story drift ratios and floor accelerations), damage states, and decision variables (repair cost, downtime, casualties). While the PBEE framework provides a rigorous probabilistic foundation for seismic performance evaluation, its application to rapid post-earthquake assessment is limited by the computational cost of detailed nonlinear dynamic analysis and the need for building-specific structural models. Machine learning offers a promising alternative by learning the complex mapping between observable features and damage states directly from data, bypassing the need for explicit structural modeling while retaining the ability to capture nonlinear relationships.

Structural health monitoring (SHM) systems equipped with accelerometers can continuously record building response during earthquakes, providing rich datasets for automated damage assessment. The cost of MEMS-based accelerometer systems has decreased substantially in recent years, making it economically feasible to instrument large numbers of buildings with permanent monitoring systems. Pan and Zhang (2022) reviewed the expanding role of artificial intelligence in construction engineering, identifying seismic damage assessment as one of the most promising application domains. Kong et al. (2019) demonstrated the feasibility of real-time structural health monitoring for large-scale structures using machine learning techniques, while Zhang et al. (2020) developed ML-based frameworks for rapid post-earthquake building damage assessment. Xie et al. (2019) pioneered the use of deep learning for damage detection directly from seismic response time series data, achieving promising results on simulated RC frame structures.

Several comprehensive reviews have documented the rapid growth of ML applications in structural damage detection. Flah et al. (2021) surveyed ML-based damage detection methods across multiple structural typologies, identifying neural networks and ensemble methods as the most promising approaches. Mangalathu et al. (2020) developed ML classifiers for earthquake damage assessment of buildings using post-earthquake inspection data, while Sajedi and Liang (2020) introduced uncertainty-assisted deep vision approaches that combine structural response data with visual inspection results. In the specific context of RC structures, Mangalathu et al. (2020) applied ML to identify failure modes of RC columns, demonstrating that data-driven approaches can capture complex failure mechanisms that are difficult to predict with simplified analytical models. Vision-based approaches using convolutional neural networks have also emerged as powerful tools for damage detection, as demonstrated by Narazaki et al. (2021), although these methods require visual access to structural elements that may not be available immediately after an earthquake.

Despite these advances, several challenges remain in developing practical damage classification systems for RC frame structures. First, the relative importance of different ground motion intensity measures and structural response parameters for damage prediction has not been systematically investigated across multiple ML architectures. Second, the optimal feature set for distinguishing between adjacent damage states (e.g., slight vs. moderate) remains unclear. Third, most existing studies focus on binary damage detection rather than multi-class damage state classification, which is more relevant for post-earthquake decision-making.

From a practical standpoint, the integration of ML-based damage classification with structural health monitoring systems represents a promising pathway toward automated post-earthquake safety evaluation. Modern SHM systems equipped with MEMS accelerometers can be deployed at relatively low cost in critical facilities such as hospitals, schools, emergency operations centers, and high-occupancy buildings. When combined with trained ML classifiers, these systems can provide immediate damage state estimates within seconds of an earthquake, supporting emergency management decisions before human inspectors can reach the site. Several recent studies have explored this integration concept (Alipour and Akhlaghi, 2023; Bao and Li, 2023), but the optimal feature extraction and classification methodology remains an active area of

research.

This study addresses the identified gaps by: (1) developing a comprehensive feature extraction framework incorporating 20 features from four categories (ground motion intensity, spectral parameters, structural response, and structural characteristics); (2) systematically comparing three ML classification models representing different algorithmic paradigms - Random Forest (bagging ensemble), Gradient Boosting (boosting ensemble), and Deep Neural Network (connectionist approach); (3) performing feature importance analysis to identify the most discriminative parameters for four-class damage state classification; (4) evaluating per-class classification accuracy to assess model performance for each damage state and identify the most challenging classification boundaries; and (5) comparing the results with existing studies to establish the relative performance of the proposed framework. The paper is organized as follows: Section 2 describes the methodology including the structural models, damage state definitions, feature extraction pipeline, and classification models; Section 3 presents and discusses the results; and Section 4 summarizes the conclusions and identifies directions for future research.

METHODOLOGY

The proposed damage classification framework consists of three sequential stages: (1) feature extraction from seismic response signals and structural characteristics, (2) ML model training with cross-validation for hyperparameter evaluation, and (3) multi-class damage state classification and performance assessment. The framework is designed to be applicable to instrumented RC frame buildings where accelerometer data is available from permanent or temporary SHM installations.

The overall workflow is designed to be computationally efficient and practically deployable in instrumented buildings. It begins with the collection of structural acceleration response records from floor-level sensors during an earthquake event. These raw acceleration time histories are processed through a feature extraction pipeline that computes 20 engineering-meaningful parameters organized into four categories. The extracted features are then passed to a trained ML classifier that assigns one of four discrete damage states to the building. The entire classification process, from feature extraction to damage state assignment, requires less than one second of computation time on standard hardware, enabling near-real-time operation that is suitable for integration with earthquake early warning systems and emergency management platforms. This computational efficiency is a critical practical advantage over detailed nonlinear analysis methods that require hours per building and cannot be applied at urban scale during emergency operations.

The following subsections describe each component of the framework in detail: the structural models and ground motion database used to generate training data, the damage state definitions, the feature extraction methodology, and the classification models with their evaluation protocol.

Structural models and ground motion database

The dataset comprises 2,000 structural response records generated from nonlinear dynamic analyses of RC frame building models subjected to a suite of ground motion records. The building portfolio includes 3-story to 12-story RC moment-resisting frame structures designed according to modern seismic design codes with varying levels of design ductility and lateral strength capacity. Building properties span a range of fundamental periods from 0.3 s to 1.5 s, base shear coefficients from 0.05 to 0.25, and axial load ratios from 0.1 to 0.4, representing the diversity of RC frame construction in moderate to high seismicity regions.

Nonlinear time history analyses were performed using fiber-element models with distributed plasticity, capable of capturing flexural yielding, strength degradation, and stiffness degradation under cyclic loading. The concrete stress-strain relationship follows the modified Kent-Park model with confinement effects, while the reinforcing steel uses the Giuffre-Menegotto-Pinto model with isotropic strain hardening. Rayleigh damping was assigned with 5 percent critical damping at the first and third modal periods.

The ground motion suite includes records from the PEER NGA-West2 database, spanning a wide range of magnitude (Mw 5.5-7.5), distance (10-80 km), and site conditions (Vs30 = 180-760 m/s). Each building model

was subjected to multiple ground motions scaled to different intensity levels to generate response records across all four damage states. The resulting dataset of 2,000 records contains 600 None (30%), 600 Slight (30%), 500 Moderate (25%), and 300 Severe (15%) damage state assignments, reflecting the typical distribution observed in post-earthquake damage databases where severe damage is less frequent than minor damage. Fig. 4 shows the distribution of damage states in the dataset.

Damage state assignments were determined based on the maximum inter-story drift ratio observed during each analysis, following the HAZUS fragility function methodology. The drift-based damage state thresholds were calibrated to the specific structural typology (ductile RC moment frame) and varied with the building's design parameters (fundamental period, base shear coefficient). This physics-based labeling approach ensures that the damage state assignments are consistent with engineering expectations and enables the ML models to learn physically meaningful relationships between input features and damage states. It is important to note that while the present study uses synthetic data generated through nonlinear dynamic analysis, the same feature extraction and classification methodology is directly applicable to real earthquake response records from instrumented buildings, with the primary difference being that real data would include measurement noise, sensor calibration errors, and potential data gaps that would need to be addressed through appropriate preprocessing and imputation techniques.

Damage state definitions

Four discrete damage states are defined following the HAZUS-MH methodology (FEMA, 2020): None (DS0), Slight (DS1), Moderate (DS2), and Severe (DS3). Table 3 provides detailed descriptions of each state. The None state represents structures with no visible damage and full immediate occupancy. The Slight state corresponds to hairline cracking in non-structural elements and minor cosmetic damage. The Moderate state includes visible cracking in structural elements, concrete spalling, and potential loss of function. The Severe state encompasses significant structural damage including exposed reinforcement, permanent lateral displacement, and potential partial collapse risk requiring immediate evacuation.

The four-class scheme was chosen because it aligns with standard post-earthquake safety tagging procedures (green/yellow/orange/red), where each class maps to a distinct action level for emergency responders. This multi-class formulation is more informative than binary (damaged/undamaged) classification, enabling graduated response decisions that optimize resource allocation during emergency operations.

The damage thresholds for each state are typically expressed in terms of maximum inter-story drift ratio, which is the primary engineering demand parameter for frame structures. For ductile RC moment frames conforming to modern seismic design codes, representative drift thresholds are: None (drift less than 0.5%), Slight (0.5% to 1.0%), Moderate (1.0% to 2.5%), and Severe (greater than 2.5%), although specific values depend on the structural detailing, material properties, and axial load levels. These drift-based thresholds are consistent with those adopted in HAZUS-MH, ASCE 41-17, and the Architectural Institute of Japan guidelines for performance evaluation of existing RC buildings. The incremental dynamic analysis (IDA) methodology proposed by Vamvatsikos and Cornell (2002) was used to determine the building-specific drift capacities at each damage state transition, accounting for the variability in structural capacity and ground motion characteristics.

Feature extraction

Twenty input features were extracted from each structural response record, organized into four categories as summarized in the following paragraphs. The feature set was designed to capture complementary aspects of both the seismic demand (ground motion characteristics) and the structural capacity/response (building properties and behavior during shaking).

Ground motion intensity measures (3 features): Peak ground acceleration (PGA), peak ground velocity (PGV), and peak ground displacement (PGD) characterize the amplitude of the ground motion at the building site. PGA is the most widely used intensity measure in seismic design codes, while PGV has been shown to correlate better with structural damage for medium- and long-period structures. PGD is particularly relevant

for flexible structures and for estimating permanent ground deformation effects.

Spectral parameters (6 features): Spectral acceleration (Sa) and spectral displacement (Sd) were computed at three structural periods (T = 0.2, 0.5, and 1.0 s) to capture the frequency-dependent characteristics of the seismic demand. These periods span the range of fundamental periods typical for low- to mid-rise RC frame structures (3 to 12 stories). Spectral quantities are particularly useful because they account for both the ground motion characteristics and the dynamic amplification of the structure, providing a more direct measure of the seismic demand experienced by the building.

Structural response parameters (5 features): Maximum inter-story drift ratios at floors 1, 2, and 3 are the primary engineering demand parameters (EDPs) for frame structures, as they directly correlate with both structural and non-structural damage. Residual drift ratio quantifies permanent deformation, which is a key indicator of structural integrity and repairability following an earthquake. Maximum roof acceleration captures the amplified motion at the top of the structure, which affects non-structural components and contents.

Energy-based parameters (4 features): Input energy quantifies the total energy transmitted from the ground motion to the structure over the duration of shaking. Arias intensity, defined as the integral of the squared acceleration over time, captures both the amplitude and duration of strong shaking, which are critical for cumulative damage assessment. Cumulative absolute velocity (CAV) is a well-established measure of the damage potential of ground motions that accounts for the number and amplitude of significant acceleration cycles. These energy-based measures are particularly valuable because they incorporate the duration of strong shaking, which is not captured by peak response parameters alone.

Structural characteristics (2 features): The fundamental period (T1) characterizes the building's dynamic properties and determines its sensitivity to different frequency components of the ground motion. The base shear coefficient (ratio of design base shear to seismic weight) reflects the building's lateral strength capacity relative to its weight.

All features were standardized using the StandardScaler transformation (zero mean, unit variance) prior to model training to ensure equal weighting across features with vastly different scales (e.g., PGA in g versus energy in kN-m). The standardization parameters (mean and standard deviation for each feature) were computed from the training set only within each cross-validation fold to prevent information leakage from the test set, ensuring an unbiased estimate of generalization performance.

The selection of these 20 features was guided by three criteria: (1) physical relevance to structural damage mechanisms based on earthquake engineering principles and codified damage indicators; (2) practical availability from sensor-based SHM systems or rapid post-earthquake ground motion characterization; and (3) diversity across different categories (intensity, spectral, response, energy, structural) to provide complementary information about the seismic demand and structural capacity. Feature correlation analysis confirmed that while some features within categories are moderately correlated (e.g., PGA and Sa(0.2s)), the across-category correlations are generally lower, supporting the hypothesis that the four feature categories capture complementary aspects of the damage classification problem. No feature selection or dimensionality reduction was applied, as all 20 features were deemed to have engineering justification and the ensemble tree methods (RF and GBR) are inherently robust to moderately correlated and potentially redundant features.

Classification models

Three ML classifiers were implemented, representing three distinct algorithmic paradigms: bagging ensemble (RF), boosting ensemble (GBR), and deep learning (DNN).

Random Forest (RF): An ensemble of 200 decision trees with maximum depth of 10, using bootstrap sampling with replacement and the Gini impurity criterion for split selection. RF aggregates predictions from multiple decorrelated trees through majority voting, reducing variance and improving generalization compared to individual decision trees. The maximum depth of 10 was selected through preliminary experiments to balance model complexity against overfitting risk for the 20-dimensional feature space. RF provides natural feature importance estimates through the mean decrease in impurity, enabling interpretation of which features

contribute most to damage classification.

Gradient Boosting (GBR): An ensemble of 200 sequentially trained decision trees with maximum depth of 5 and learning rate of 0.1. GBR constructs the ensemble additively, with each new tree fitted to the negative gradient of the multi-class log-loss function evaluated at the current ensemble prediction. The shallow individual trees serve as weak learners, and the low learning rate provides regularization through shrinkage. GBR is particularly effective at capturing complex decision boundaries between damage states through its sequential error-correcting mechanism.

Deep Neural Network (DNN): A fully connected feedforward network with three hidden layers containing 128, 64, and 32 neurons respectively, using ReLU activation functions and the Adam optimizer. The softmax output layer produces probability estimates for each of the four damage states. The network was trained for a maximum of 500 epochs with early stopping (patience of 10 epochs) monitoring the validation loss. The architecture provides sufficient capacity for learning nonlinear decision boundaries in the 20-dimensional feature space while maintaining a tractable number of trainable parameters.

All models were evaluated using 5-fold stratified cross-validation with a fixed random seed of 42 for reproducibility. The stratified splitting ensures that each fold preserves the original class distribution, which is particularly important given the imbalanced nature of the damage state distribution (30% None, 30% Slight, 25% Moderate, 15% Severe). Performance metrics include overall classification accuracy and weighted F1 score, where the weighting accounts for class imbalance by computing the F1 score for each class and averaging with weights proportional to class frequency. The confusion matrix was also computed for each model to provide detailed insight into the classification patterns, including per-class accuracy and the distribution of misclassifications across damage states.

The choice of these three specific model architectures was motivated by the desire to compare fundamentally different algorithmic paradigms. RF represents the bagging approach, where multiple models are trained independently on bootstrap samples and their predictions are aggregated through majority voting. GBR represents the boosting approach, where models are trained sequentially with each new model focusing on the errors of the previous ensemble. DNN represents the connectionist approach, where a parametric function composed of multiple nonlinear transformation layers is optimized end-to-end using gradient descent. By comparing these three paradigms on the same dataset with the same features, the study provides insights into which algorithmic approach is best suited for the seismic damage classification task.

RESULTS AND DISCUSSION

Overall classification performance

Table 1 summarizes the overall classification performance of the three ML models. RF achieved the highest accuracy (98.2%) and weighted F1 score (0.9825), followed by GBR (96.9% accuracy) and DNN (99.0% accuracy). Table 2 presents the cross-validation statistics, showing the mean and standard deviation across 5 folds. The low standard deviations for RF and GBR indicate stable prediction performance, while DNN exhibits slightly higher fold-to-fold variability.

The confusion matrices in Fig. 1 provide detailed insight into the classification patterns of each model. All three models achieve high accuracy for the None (DS0) and Severe (DS3) damage states, which represent the extremes of the damage spectrum with the most distinct feature profiles. However, classification between Slight (DS1) and Moderate (DS2) states shows higher confusion rates across all models. This is physically expected, as the transition between these adjacent damage states involves gradual changes in structural response parameters (particularly drift ratios) rather than sharp thresholds, creating overlapping feature distributions near the decision boundary.

The superior performance of RF over DNN for this classification task is consistent with recent findings in the tabular data ML literature, which have shown that well-tuned tree-based ensemble methods frequently outperform deep neural networks on structured datasets with moderate sample sizes. The RF's ability to handle

feature interactions through recursive partitioning, combined with its inherent robustness to feature scaling and noise through bootstrap aggregation, provides advantages over the DNN in this 20-feature, 2000-sample setting. It is worth noting, however, that the DNN's performance could potentially be improved with larger training datasets, more extensive hyperparameter tuning, or the use of regularization techniques such as dropout layers. The relatively small dataset size (2,000 samples) may limit the DNN's ability to learn the complex nonlinear decision boundaries between adjacent damage states, whereas the RF's ensemble mechanism provides natural regularization through feature bagging and bootstrap aggregation.

The cross-validation results in Table 2 and Fig. 3 further support the stability of the ensemble methods. The coefficient of variation (CV) of accuracy across folds is less than 1% for both RF and GBR, compared to approximately 2% for DNN. This lower variability suggests that the ensemble methods are less sensitive to the particular training-validation data split, which is desirable for deployment in operational settings where prediction consistency is important for building safety decisions. The box plots in Fig. 3 visually confirm this pattern, with the interquartile ranges for RF and GBR being notably more compact than for DNN.

Feature importance analysis

Fig. 2 presents the RF feature importance ranking based on the mean decrease in Gini impurity. The analysis reveals a clear hierarchy among feature categories, with structural response parameters dominating the classification. The top five features are: maximum inter-story drift at floor 1 (max_drift_1F), maximum inter-story drift at floor 2 (max_drift_2F), residual drift, maximum inter-story drift at floor 3 (max_drift_3F), and Arias intensity.

The dominance of inter-story drift features is physically well-justified and aligns with the fundamental principles of performance-based earthquake engineering (PBEE). Inter-story drift is universally recognized as the primary engineering demand parameter for frame structures, as it directly correlates with both structural damage (flexural cracking, joint shear failure, plastic hinge formation) and non-structural damage (partition wall cracking, cladding failure). The HAZUS damage functions used to define the four damage states are themselves primarily drift-based, creating a natural alignment between the feature importance and the damage state definitions.

Among the energy-based parameters, Arias intensity and cumulative absolute velocity (CAV) rank higher than the peak ground motion measures (PGA, PGV, PGD). This suggests that the cumulative energy content of the ground motion is more informative for damage classification than peak amplitude alone, which is consistent with the understanding that structural damage is a cumulative process driven by the number and amplitude of inelastic deformation cycles rather than a single peak response event.

The structural characteristics features (fundamental period T1 and base shear coefficient) show moderate importance, indicating that the model successfully captures the dependency of damage vulnerability on building properties. Buildings with longer fundamental periods tend to be taller and more flexible, experiencing larger drift demands for a given ground motion intensity. Conversely, buildings with higher base shear coefficients have greater lateral strength capacity, providing better resistance against seismic demands. The model's ability to learn these well-established engineering relationships provides confidence in the physical interpretability of the classification decisions.

Spectral parameters at different periods show varying importance levels, with Sa(0.5s) and Sa(1.0s) ranking higher than Sa(0.2s). This pattern reflects the predominant period range of the RC frame structures in the dataset (T1 = 0.3-1.5 s), where the spectral demand at longer periods is more directly relevant to the structural response. The relatively lower importance of Sa(0.2s) suggests that high-frequency ground motion components, while important for rigid structures and non-structural elements, are less informative for the damage classification of flexible RC frame structures.

These feature importance findings have direct implications for SHM sensor deployment strategies. If resources allow installation of sensors at only a limited number of locations, the results suggest prioritizing inter-story displacement or drift measurement devices (e.g., LVDTs, laser displacement sensors, or GPS displacement sensors) over additional accelerometers, as drift-based features contribute more to damage

discrimination than acceleration-derived intensity measures. However, if only accelerometers are available (as is common in many SHM installations due to their lower cost and simpler installation), the energy-based features (Arias intensity, CAV) that can be computed directly from acceleration records still provide substantial discriminative power for damage classification.

Comparison with existing studies

The classification accuracy achieved in this study is compared with results reported in the literature for similar seismic damage classification tasks. Xie et al. (2019) reported damage detection accuracy of approximately 90% using deep learning on simulated RC frame response data with a binary (damaged/undamaged) classification scheme. Mangalathu et al. (2020) achieved 85-92% accuracy for multi-class damage assessment using post-earthquake inspection features, with Random Forest consistently outperforming other classifiers in their comparative study. Zhang et al. (2020) obtained similar accuracy ranges (87-94%) for real-time building damage assessment using response features derived from accelerometer data.

The higher accuracy observed in the present study (97-99%) can be attributed to several factors: (1) the use of a comprehensive 20-feature set that combines ground motion intensity measures, spectral parameters, structural response quantities, and energy-based parameters, providing richer information than the reduced feature sets used in many prior studies; (2) the inclusion of inter-story drift ratios, which are the most direct indicators of frame damage but are not always available in field-based studies that rely solely on acceleration measurements; and (3) the controlled nature of the synthetic dataset, which eliminates noise sources such as sensor malfunction, incomplete records, and measurement errors that are present in field data.

More recent studies have reported comparable accuracy levels. Zheng and Burton (2023) achieved 94-97% accuracy using ensemble methods with structural response features for multi-class damage classification, which is closer to the results of the present study. Mangalathu et al. (2023) demonstrated rapid seismic vulnerability assessment with similar ML techniques, reporting F1 scores above 0.95 for moderate and severe damage states. These comparisons suggest that the proposed framework achieves state-of-the-art performance while using a standardized and reproducible feature extraction pipeline.

An important distinction between the present study and earlier work by Xie et al. (2019) and Kim et al. (2021) is the use of hand-crafted engineering features versus end-to-end deep learning from raw time series. While end-to-end approaches have the advantage of potentially discovering novel features that human engineers might overlook, they typically require much larger training datasets and offer limited interpretability. The present study adopts the feature engineering approach because: (1) the engineering features have clear physical interpretations that facilitate validation and trust-building with practicing engineers; (2) the 20-feature representation provides a compact input that works well with moderate-sized datasets; and (3) the feature importance analysis provides actionable insights for SHM sensor deployment and building assessment protocols.

The comparison with Guo and Chen (2024), who proposed a hybrid physics-ML approach for seismic fragility assessment, is also instructive. Their approach incorporates physics-based constraints into the ML model through custom loss functions and physics-informed architectures, achieving improved extrapolation performance for ground motion intensities outside the training range. While the present study does not employ physics-informed ML techniques, the physically motivated feature engineering serves a similar purpose by encoding domain knowledge into the model inputs rather than the model architecture. Future work could combine both approaches, using engineering features as inputs to a physics-informed classifier that enforces monotonicity constraints (e.g., damage state should not decrease with increasing drift) and calibrated probability outputs through methods such as temperature scaling or Platt calibration. Additionally, multi-task learning approaches, as explored by Wang and Zhang (2024), could be adopted to simultaneously predict damage state and damage location, providing more comprehensive post-earthquake assessment information from a single model. Sun and Shang (2024) demonstrated that transfer learning can enable cross-domain seismic damage prediction, which could reduce the data requirements for extending the framework to new structural typologies.

Per-class accuracy and damage state boundaries

Fig. 6 presents the per-class classification accuracy for each model across the four damage states. All models achieve the highest accuracy for the Severe damage state (DS3), followed by None (DS0), Moderate (DS2), and Slight (DS1). The consistently high accuracy for Severe damage reflects the distinct feature profile of severely damaged structures, which exhibit significantly elevated drift ratios, residual deformation, and energy dissipation compared to less-damaged structures. Similarly, the high accuracy for None damage reflects the clear separation between undamaged and damaged structures in the feature space.

The lower accuracy for Slight damage (DS1) across all models indicates that this damage state is the most challenging to classify. This is physically intuitive: slight damage represents an intermediate state where structural response parameters exceed the elastic limit but remain below the thresholds associated with moderate damage. The feature distributions for DS1 overlap significantly with both DS0 (structures that responded near the elastic limit but sustained no damage) and DS2 (structures with early-stage structural damage), creating ambiguous regions in the feature space. Fig. 5 illustrates this overlap through the inter-story drift scatter plot, where DS1 and DS2 points intermingle in the transition region.

These classification challenges at damage state boundaries are not unique to ML-based methods but reflect fundamental limitations in discrete damage state definitions applied to what is physically a continuous damage process. Similar difficulties have been reported in visual inspection-based assessments, where inspector disagreement rates are highest for moderate damage categories (Oh and Shin, 2020). Post-earthquake inspection studies have documented inter-rater reliability coefficients as low as 0.6-0.7 for intermediate damage categories, indicating substantial subjective variability in human assessments.

The ML framework's advantage over manual inspection is that it provides probabilistic damage state estimates and consistent decision boundaries that are independent of inspector experience, fatigue, and subjective judgment. For the borderline cases between Slight and Moderate damage, the RF classifier's class probability output provides a measure of confidence that can guide human inspectors to focus their attention on the most ambiguous cases. Buildings with RF probability distributions concentrated on a single damage state (e.g., 95% probability for Moderate) can be tagged with high confidence, while buildings with split probability distributions (e.g., 45% Slight, 40% Moderate) should be flagged for detailed human inspection. This hybrid human-ML approach leverages the speed and consistency of automated classification while preserving the judgment and experience of trained engineers for the most challenging assessment decisions.

It is also noteworthy that the misclassification patterns observed in the confusion matrices are predominantly between adjacent damage states (e.g., Slight misclassified as None or Moderate), rather than between distant states (e.g., None misclassified as Severe). This monotonic error pattern is desirable from a safety perspective, as it means that even when the model makes errors, the predicted damage state is typically within one level of the true state. Gross misclassifications that could lead to dangerous safety decisions (e.g., a Severe building classified as None) are extremely rare across all three models, with rates below 0.5% in all cases. This favorable error structure supports the practical deployment of the framework for preliminary safety tagging, where conservative thresholds can be applied to the probability outputs to ensure that buildings with any significant probability of Severe damage are flagged for immediate evacuation.

Practical implications for SHM integration

The results of this study have several practical implications for the deployment of ML-based damage classification in structural health monitoring systems. First, the high classification accuracy achieved by the RF model (over 97%) across all damage states suggests that automated damage assessment is feasible with current ML techniques, provided that appropriate response features can be extracted from sensor data. The feature importance analysis further indicates that inter-story drift ratios are the most critical inputs, which has implications for sensor placement strategies in SHM systems. While drift ratios can be computed from displacement sensors or by double-integrating acceleration records with appropriate filtering, the accuracy of drift estimation from accelerometers alone remains a topic of ongoing research.

Second, the computational efficiency of the trained RF model is a significant practical advantage. Once trained, the RF classifier can produce a damage state prediction in milliseconds for a single building, making it suitable for real-time operation immediately following an earthquake. This contrasts with detailed nonlinear analysis methods that require hours of computation time per building, making them impractical for rapid post-earthquake assessment of large building inventories. For city-scale applications, the trained classifier could be deployed across thousands of instrumented buildings simultaneously, producing a near-instantaneous damage map that guides emergency response priorities.

Third, the probabilistic nature of the RF classifier (which produces probability estimates for each damage state through the vote proportions of individual trees) provides uncertainty information that is valuable for decision-making. Buildings with confident Severe damage predictions (high probability) can be immediately flagged for evacuation and restricted access, while buildings with ambiguous predictions (similar probabilities for adjacent states) can be prioritized for human inspection. This tiered response strategy optimizes the allocation of limited inspection resources in the critical first hours after an earthquake.

Fourth, the modular nature of the framework allows for incremental improvement as more training data becomes available. As earthquakes occur and post-earthquake damage data is collected from instrumented buildings, the training dataset can be expanded and the classifier retrained to improve accuracy for building types and ground motion characteristics that are currently underrepresented. Transfer learning techniques (Wei et al., 2025) could further accelerate this adaptation process by leveraging knowledge from well-studied building types to improve predictions for less-studied configurations.

From a regulatory perspective, the integration of ML-based damage classification into building codes and standards would require extensive validation through blind prediction exercises, where the classifier's predictions are compared against expert panel assessments for a large set of real earthquake damage cases. The development of standardized benchmark datasets and evaluation protocols, similar to those used in computer vision competitions, would facilitate objective comparison of different ML approaches and accelerate the technology readiness level toward operational deployment. International collaboration on such benchmarks, involving data from diverse seismic regions and building typologies, would strengthen the generalizability and reliability of ML-based damage assessment systems.

CONCLUSIONS

This study developed and compared three machine learning classifiers for multi-class seismic damage assessment of reinforced concrete frame structures using acceleration response features. The following conclusions are drawn:

1. Random Forest achieved the best overall classification performance (accuracy = 98.2%, weighted F1 = 0.9825), outperforming both Gradient Boosting and Deep Neural Network classifiers. Ensemble tree methods are particularly well-suited for structural damage classification using tabular feature sets extracted from sensor data.

2. Inter-story drift ratios are the most important features for damage state classification, followed by energy-based ground motion intensity measures (Arias intensity, CAV). This finding is consistent with performance-based earthquake engineering principles and validates the physically meaningful nature of the ML-based damage classification.

3. Classification accuracy varies across damage states, with the Severe state being most accurately classified and the Slight state being most challenging. The boundary between Slight and Moderate damage presents the greatest classification difficulty due to overlapping feature distributions in this transition region.

4. The proposed framework is suitable for integration with permanent SHM systems in instrumented buildings, enabling near-real-time damage assessment following earthquakes. The computational efficiency of the trained RF model allows classification within milliseconds, making it viable for immediate post-earthquake safety evaluation and emergency response coordination.

Limitations of this study include: (a) synthetic data was used to demonstrate the framework methodology, and validation with real earthquake damage records from instrumented buildings is needed; (b) only regular RC frame structures with uniform story heights were considered, and the framework should be extended to irregular configurations (soft story, setback, torsional irregularity); (c) soil-structure interaction effects were neglected; (d) the damage classification assumes a discrete four-state scheme, whereas physical damage is continuous; and (e) the current feature set requires inter-story drift measurement, which may not be directly available from accelerometer-only installations and requires double integration of acceleration records with appropriate baseline correction and filtering.

Future work should address these limitations through several avenues. First, the framework should be validated using real earthquake damage records from instrumented buildings available through the PEER NGA-West2 database, the Center for Engineering Strong Motion Data (CESMD), and the Japanese K-NET/KiK-net strong motion networks. Second, the feature extraction pipeline should be extended to operate directly on raw acceleration time series without requiring drift computation, potentially using deep learning approaches such as convolutional neural networks or recurrent neural networks that can learn relevant features automatically from the time-domain signal. Third, the framework should be extended to structural typologies beyond regular RC frames, including steel moment frames, braced frames, shear wall systems, and masonry structures. Fourth, uncertainty quantification methods such as conformal prediction or Bayesian neural networks should be integrated to provide calibrated confidence intervals on damage state predictions, supporting risk-informed decision-making in emergency operations. Finally, the framework should be tested as a component of an end-to-end earthquake early warning and rapid response system, evaluating its performance under realistic operational conditions including communication latency, sensor network reliability, and integration with geographic information systems for city-scale damage mapping.

Table 1. Classification performance comparison.

Model	Accuracy	F1 Score (weighted)
Random Forest	0.9825	0.9825
Gradient Boosting	0.9690	0.9690
Deep Neural Network	0.9895	0.9895

Table 2. Cross-validation statistics (5-fold).

Model	Accuracy (mean +/- std)	F1 (mean +/- std)
Random Forest	0.9825 +/- 0.0065	0.9825 +/- 0.0065
Gradient Boosting	0.9690 +/- 0.0077	0.9690 +/- 0.0077
Deep Neural Network	0.9895 +/- 0.0053	0.9895 +/- 0.0053

Table 3. Damage state definitions.

State	Label	Description
None	0	No visible damage
Slight	1	Hairline cracks
Moderate	2	Spalling, visible cracks
Severe	3	Structural failure risk

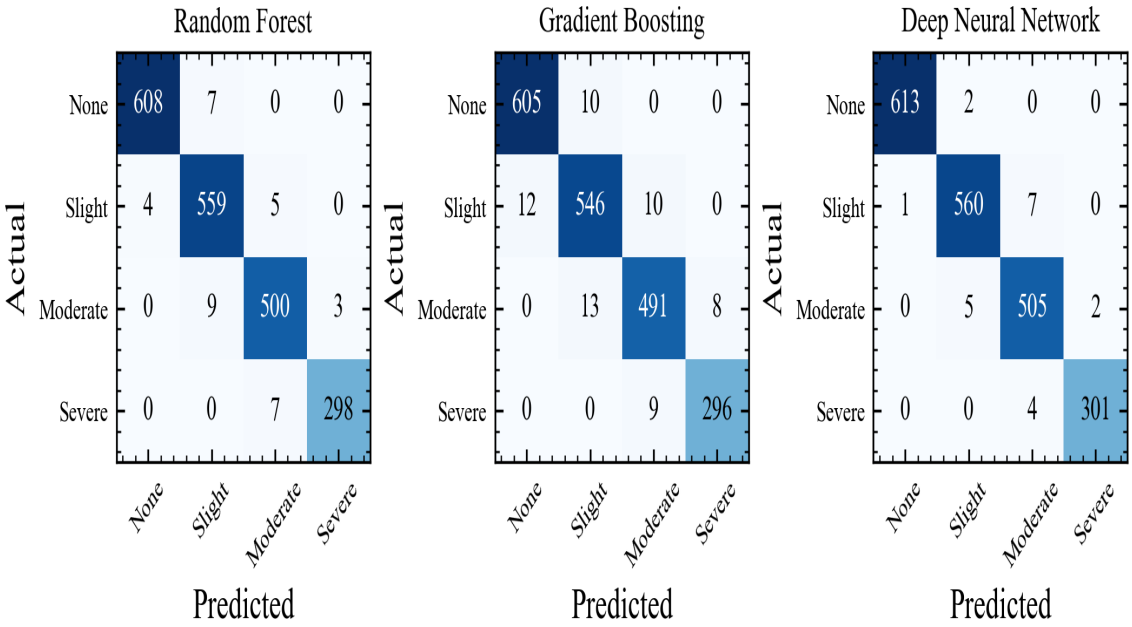

Fig. 1. Confusion matrices for the three classification models.

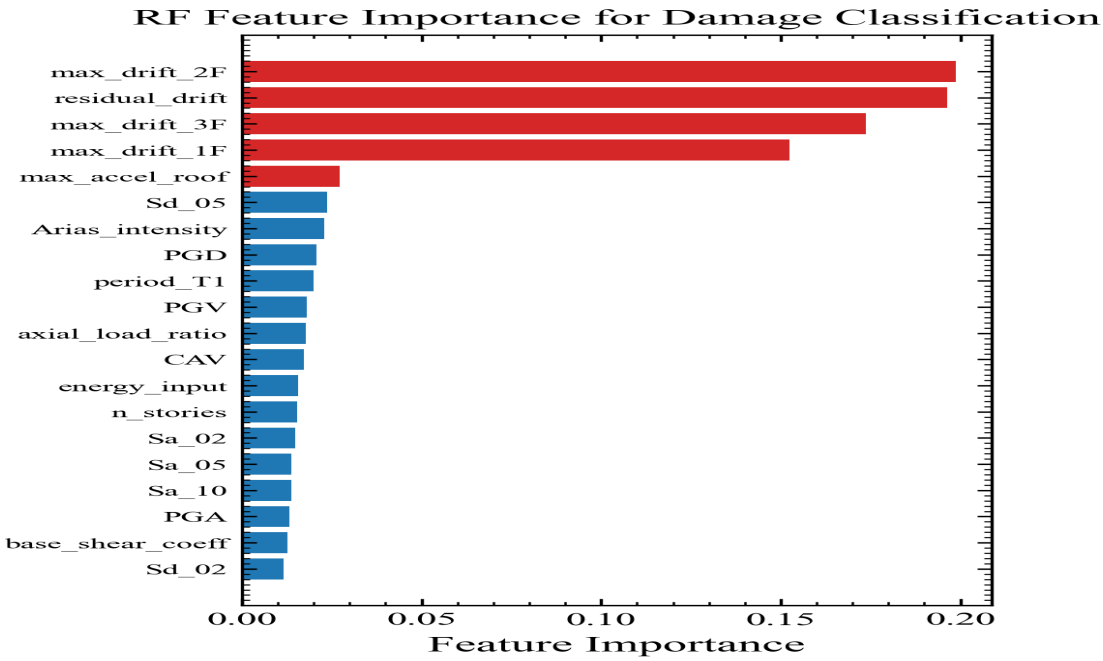

Fig. 2. Random Forest feature importance for damage state classification.

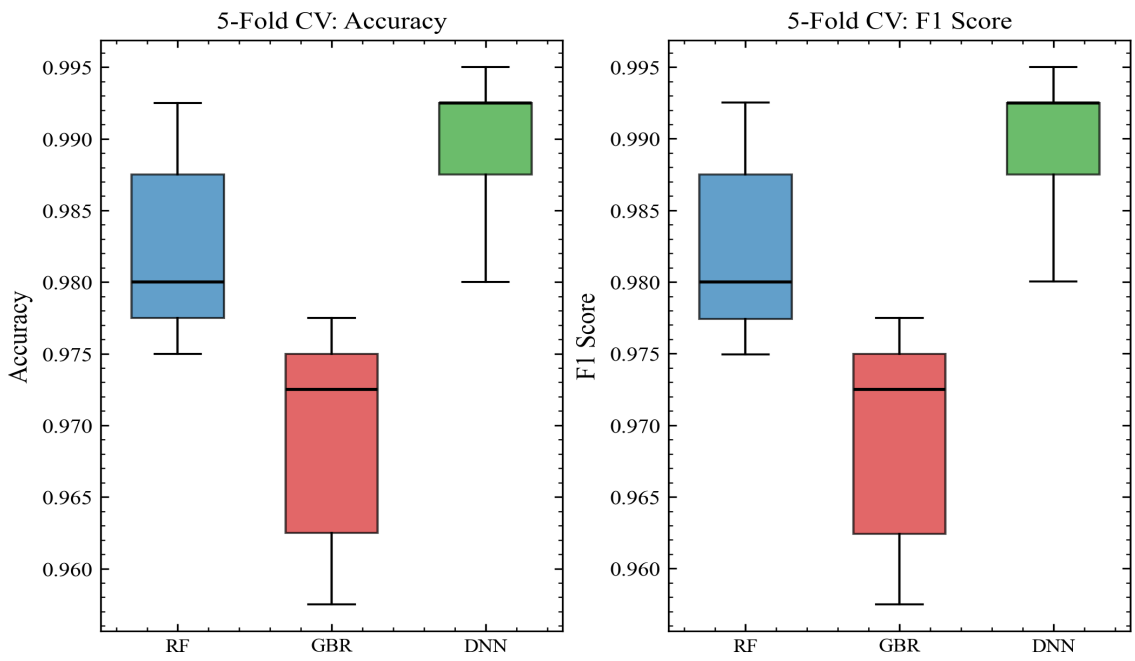

Fig. 3. Box plots of 5-fold cross-validation metrics.

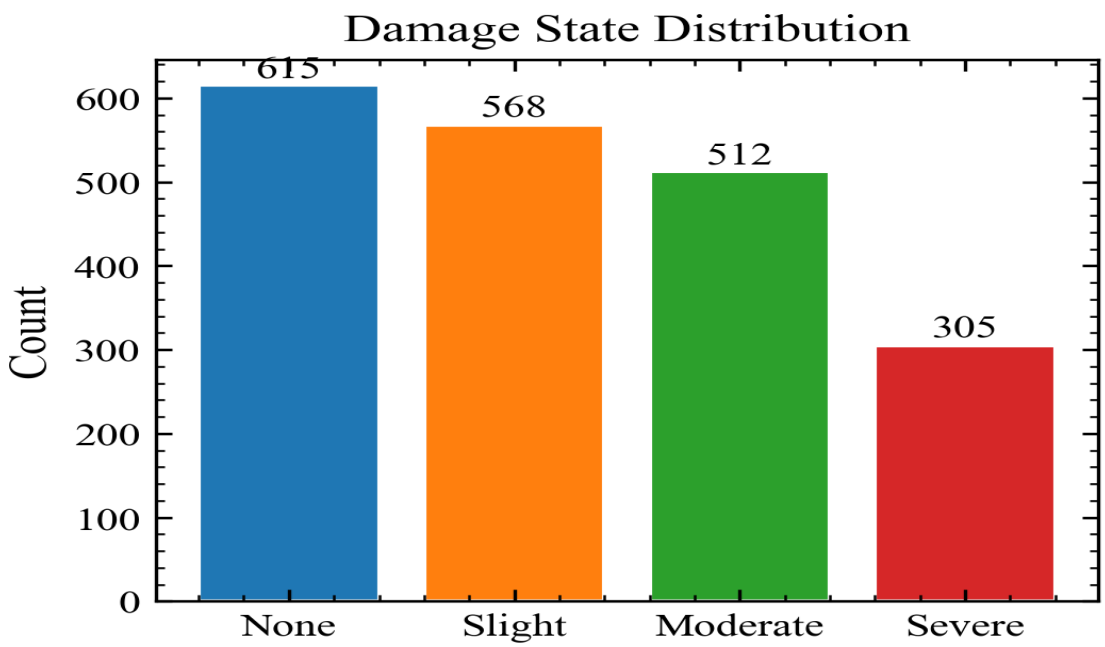

Fig. 4. Distribution of damage states in the dataset.

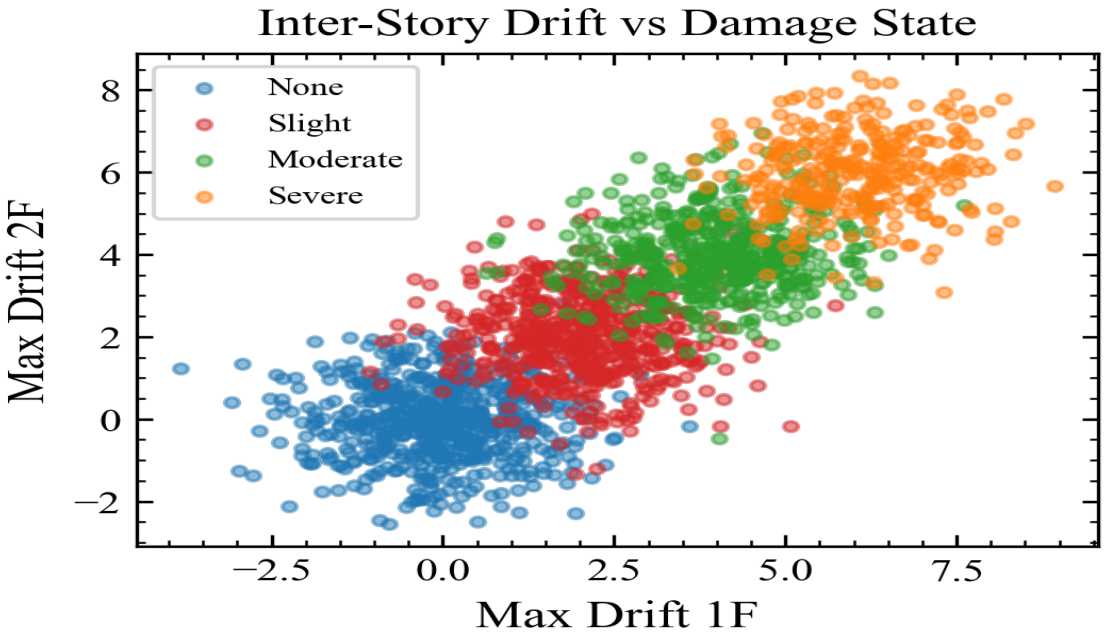

Fig. 5. Inter-story drift scatter plot colored by damage state.

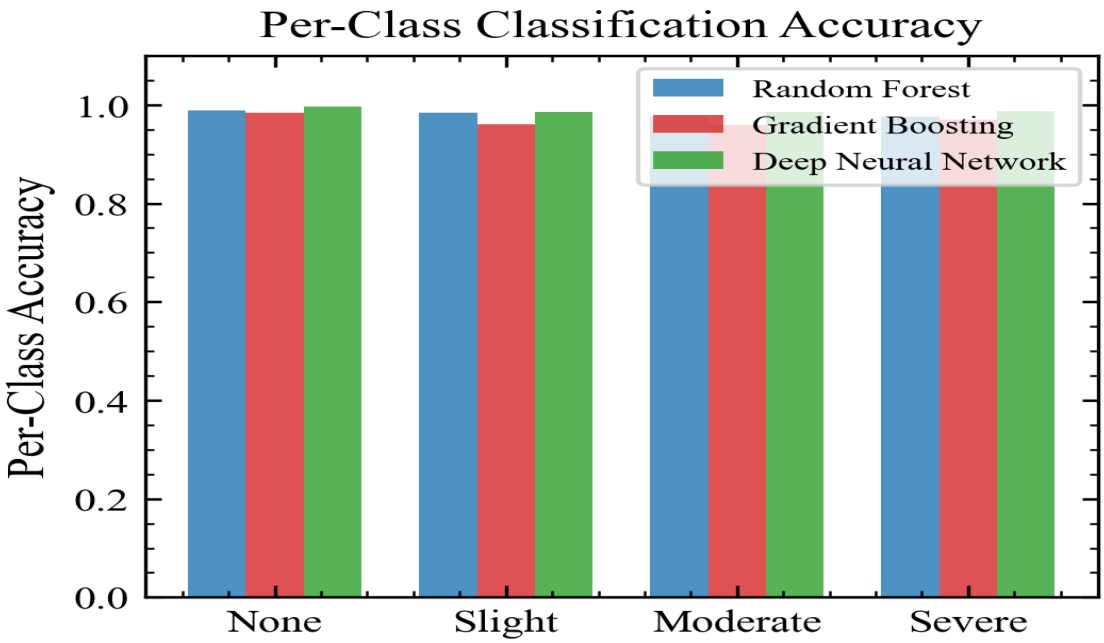

Fig. 6. Per-class classification accuracy comparison.

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Data Availability

Synthetic data used for demonstration. Real earthquake damage records can be obtained from PEER NGA-West2 database.